CANONICAL BASELINE PROJECT ENG1M-0000001

MODULAR MICRO-CELL ARCHITECTURE WITH PREDICTIVE EDGE ORCHESTRATION

Industrial Engineering Turnaround for High-Throughput Cold-Chain Fulfillment

EXECUTIVE SUMMARY: A 250,000 sq.ft cold-chain fulfillment center experiencing severe throughput starvation (baseline 1,190 units/day) faced catastrophic volume bottlenecks under seasonal demand spikes (+30%). We engineered the Modular Micro-Cell Architecture with Predictive Edge Orchestration (MM-PEO). By decoupling monolithic mechanical conveyors into self-balancing autonomous cells, this CAD $866,000 turnaround delivers +24.5% net output (1,482 units/day), slashes unplanned stoppages by 38.0%, and achieves capital payback in 8.4 months with a 10-year NPV of CAD $3.84M.

CAD $866K
INITIAL INVESTMENT
8.4-Month Payback

+24.5%
THROUGHPUT GAIN
1,482 Units / Day Target

+$3.84M
10-YEAR NPV
44.2% Internal Rate of Return

-38.0%
UNPLANNED DOWNTIME
Closed-Loop Telemetry